Waves in Supersonic Flow

Compressible Fluid Mechanics, AA 504, Autumn 2026, UW

Assignment: Homework #4

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```
clear % Clears workspace
clc % Clears command window
close all % Closes all figures
```

Problem 1. Shock Wave Propagation

Problem Statement

A Mach 3 shock wave propagates to the right along a shock tube and then collides with the closed end of the tube. The initial pressure of the gas is p1. Draw the x-t diagram. What is the pressure on the end of the tube after the wave arrives? (*Hint:* The boundary condition of zero speed at the end of the tube can be satisfied by adding a virtual wave of identical strength propagating to the left.)

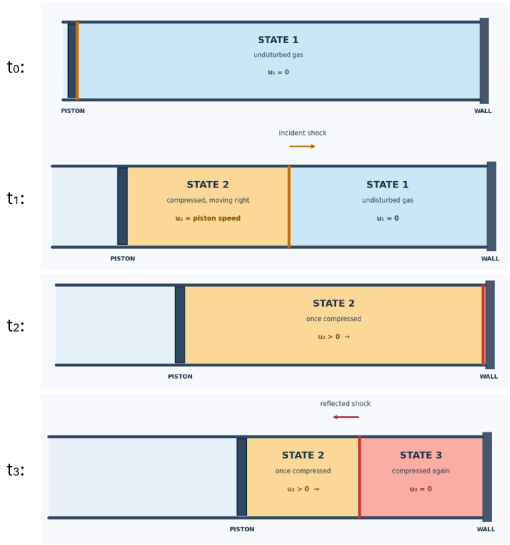
Analysis

[*pending*]

1. Understand the physical situation

Piston-generated wave analogy

```
imshow("HW4_Prob1.png")
```



at

t_0: Piston pushes right with a speed u_p, impulsively, generating a shock wave.

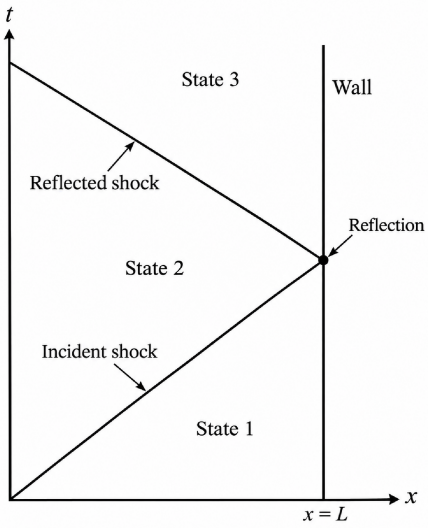
t_1: The incident shock outruns the piston, propagates to the right with a speed u_s - to where it will contact undisturbed gas in State 1, and compresses the gas leaving it in State 2.

t_2: The shock wave reaches the left wall.

t_3: The shock reflects to the left with a speed u_r, compresses the gas again where it is brought to rest, leaving it in State 3. The reflected shock is moving towards the once compressed gas from State, which is still moving right with a speed $u_p = u_2$.

2. Draw the x-t diagram

```
imshow("Black-and-White Shock Reflection Diagram.png")
```



State 1: Initially stationary, undisturbed gas.

State 2: Piston generates a rightward moving shock, compressing the gas and accelerating it to u_2.

State 3: The shock reflects from the left wall, moves leftward compressing the gas again and bringing it to rest, $u_3 = 0$.

3. Calculate the pressure behind the incident shock, p_2.

Using the normal-shock pressure ratio, which is derived from conservation momentum, and the density ratio - which is derived from conservation of mass, momentum, and energy [1]:

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_s^2 - 1) \qquad (2.48a)$$

Given: $M_s = 3.0$, $\gamma_{air} = 7/5 = 1.4$

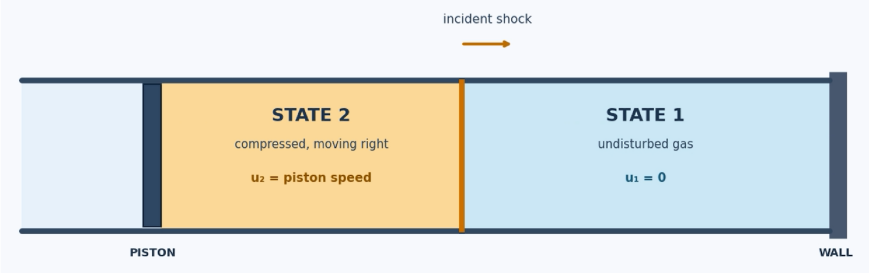
$$p_2 = 10.33 p_1$$

The pressure after the incident shock is 10.33 times its original value, but note that we are looking for the pressure at the wall **after** the shock arrives/reflects, which would be p_3.

4. Calculate the speed of the gas at state 2, u_2.

Apply conservation of mass across the shock:

```
imshow('HW4_1.2.png');
```



$\rho_2 A_2 v_2 = \rho_1 A_1 v_1$

Where,

$A_2 = A_1$

$v_1 = u_s - u_1$

$v_2 = u_s - u_2$, this is the relative velocity of the incident shock to the moving gas in State 2.

Solve for u_2:

$$u_2 = u_s\left(1 - \frac{\rho_1}{\rho_2}\right)$$

- From conservation of mass alone, we're able to solve for u_2 in terms of the density ratio, and shock speed.

Substitute the normal-shock density ratio relation [1]:

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_s^2}{2+(\gamma-1)M_s^2} \quad (2.47)$$

At: $M_s = 3.0$, $\gamma_{air} = 7/5 = 1.4$

$$u_2 = 2.222 a_1$$

We're also going to need the speed of sound at State 2, to analyze the reflected shock coming off the wall.

For a calorically perfect gas:

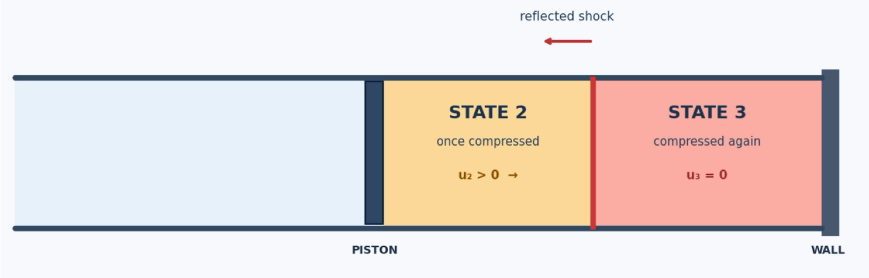
$$\frac{a_2}{a_1} = \sqrt{\frac{p_2}{p_1}\frac{\rho_1}{\rho_2}}$$

$$a_2 = 1.637 a_1$$

5. Analyze the reflected shock, M_r.

Apply the conservation of mass across the reflected shock:

```
imshow('HW4_1.4.png')
```



$$\rho_2 A_2 v_2 = \rho_3 A_3 v_3$$

Where,

$$A_2 = A_3$$

Relative velocities in the shock-fixed reference frame:

$$v_2 = -u_r + u_2$$

$$v_3 = -u_r + u_3$$

With this mass balance we're trying to express the velocity change across the reflected shock, $u_2 - u_3$, in terms of the density ratio ρ_2/ρ_3.

$$u_2 - u_3 = u_2 - u_r\left(1 - \frac{\rho_2}{\rho_3}\right)$$

Substituting the normal-shock density relation and simplifying:

$$u_2 - u_3 = \frac{2a_2}{(\gamma+1)}\frac{(M_r^2-1)}{M_r}$$

Insert previous quantities and multiply by M_r to yield a reflected shock Mach # polynomial:

$$M_r^2 - 1.629M_r - 1 = 0$$

Solve for the positive M_r root:

$$M_r = 2.1042$$

This is the reflected shock Mach # relative to the moving gas in State 2.

6. Calculate the pressure on the end of the tube after the shock arrives, p_3.

Apply the normal shock pressure ratio [1]:

$$\frac{p_3}{p_2} = 1 + \frac{2\gamma}{\gamma+1}(M_r^2-1) \quad \text{(2.48a)}$$

$$\frac{p_3}{p_2} = 5.0$$

- The reflected shock produces a pressure-jump of a factor of 5 from the pressure at State 2 to the pressure at State 3.

Combining the pressure ratios across the two shocks to find the pressure on the rigid "left" wall:

$$\frac{p_3}{p_1} = \frac{p_3}{p_2}\frac{p_2}{p_1}$$

$$p_3 = p_{wall} = 51.64p_1$$

This pressure behind the reflected shock is 51x the pressure behind the incident shock; it must be of significant strength to satisfy the boundary condition of zero speed at the end of the tube. I.e., The static pressure behind the reflected shock must be sufficiently strong to halt the incoming gas's momentum from State 2 to a complete stop.

x-t diagram:

```
%% Mach 3 Shock Reflection - x-t Diagram
% Incident shock travels right and reflects from a rigid wall.
% Horizontal axis: position x
% Vertical axis: time t
%
% Assumptions:
%   Air, gamma = 1.4
%   Initially stationary gas
```

```
%   Incident shock Mach number Ms = 3

clear; clc; close all;

%% 1. Gas and shock properties
gamma = 1.4;
Ms = 3;

a1 = 343;           % Initial speed of sound [m/s]
L = 1.0;            % Tube length [m]

% Incident shock speed (laboratory frame)
Vs = Ms*a1;

% State 2: behind incident shock
p2_p1 = 1 + 2*gamma/(gamma+1)*(Ms^2-1);

rho2_rho1 = ((gamma+1)*Ms^2)/ ...
            ((gamma-1)*Ms^2+2);

u2 = Vs*(1-1/rho2_rho1);

% Speed of sound behind incident shock
a2 = a1*sqrt(p2_p1/rho2_rho1);

%% 2. Reflected shock properties
% Reflected shock must bring the gas to rest: u3 = 0

K = (gamma+1)*u2/(2*a2);

% Solve MR - 1/MR = K
MR = (K + sqrt(K^2+4))/2;

% Reflected shock speed in laboratory frame
Vr = u2 - MR*a2;     % Negative = moving left

% Pressure behind reflected shock
p3_p2 = 1 + 2*gamma/(gamma+1)*(MR^2-1);
p3_p1 = p2_p1*p3_p2;

%% 3. Shock trajectories
% Incident shock begins at x = 0, t = 0
% Reflection occurs at x = L

tR = L/Vs;           % Time shock reaches wall
tEnd = 1.8*tR;       % Plot beyond reflection

tIncident = linspace(0,tR,200);
xIncident = Vs*tIncident;
```

```
tReflected = linspace(tR,tEnd,200);
xReflected = L + Vr*(tReflected-tR);

%% 4. Plot x-t diagram
figure('Color','w','Position',[150 100 850 650]);
hold on; box on;

% Incident shock
h1 = plot(xIncident,1e3*tIncident, ...
    'LineWidth',2.8,'Color',[0.90 0.42 0.12]);

% Reflected shock
h2 = plot(xReflected,1e3*tReflected, ...
    'LineWidth',2.8,'Color',[0.10 0.50 0.72]);

% Closed wall
h3 = xline(L,'k--','LineWidth',1.8);

% Reflection point
h4 = plot(L,1e3*tR,'ko', ...
    'MarkerFaceColor','k','MarkerSize',7);

%% 5. Region labels
text(0.75*L,0.35*1e3*tR, ...
    {'State 1','p_1, u_1 = 0'}, ...
    'HorizontalAlignment','center','FontSize',12);

text(0.30*L,0.90*1e3*tR, ...
    {'State 2','p_2 = 10.33p_1'}, ...
    'HorizontalAlignment','center','FontSize',12);

text(0.85*L,1.48*1e3*tR, ...
    {'State 3','p_3 = 51.67p_1','u_3 = 0'}, ...
    'HorizontalAlignment','center','FontSize',12);

% Reflection annotation
text(L-0.03,1e3*tR+0.04, ...
    'Reflection', ...
    'HorizontalAlignment','right', ...
    'FontSize',11,'FontWeight','bold');

%% 6. Format figure
xlabel('Position, x [m]','FontSize',13);
ylabel('Time, t [ms]','FontSize',13);

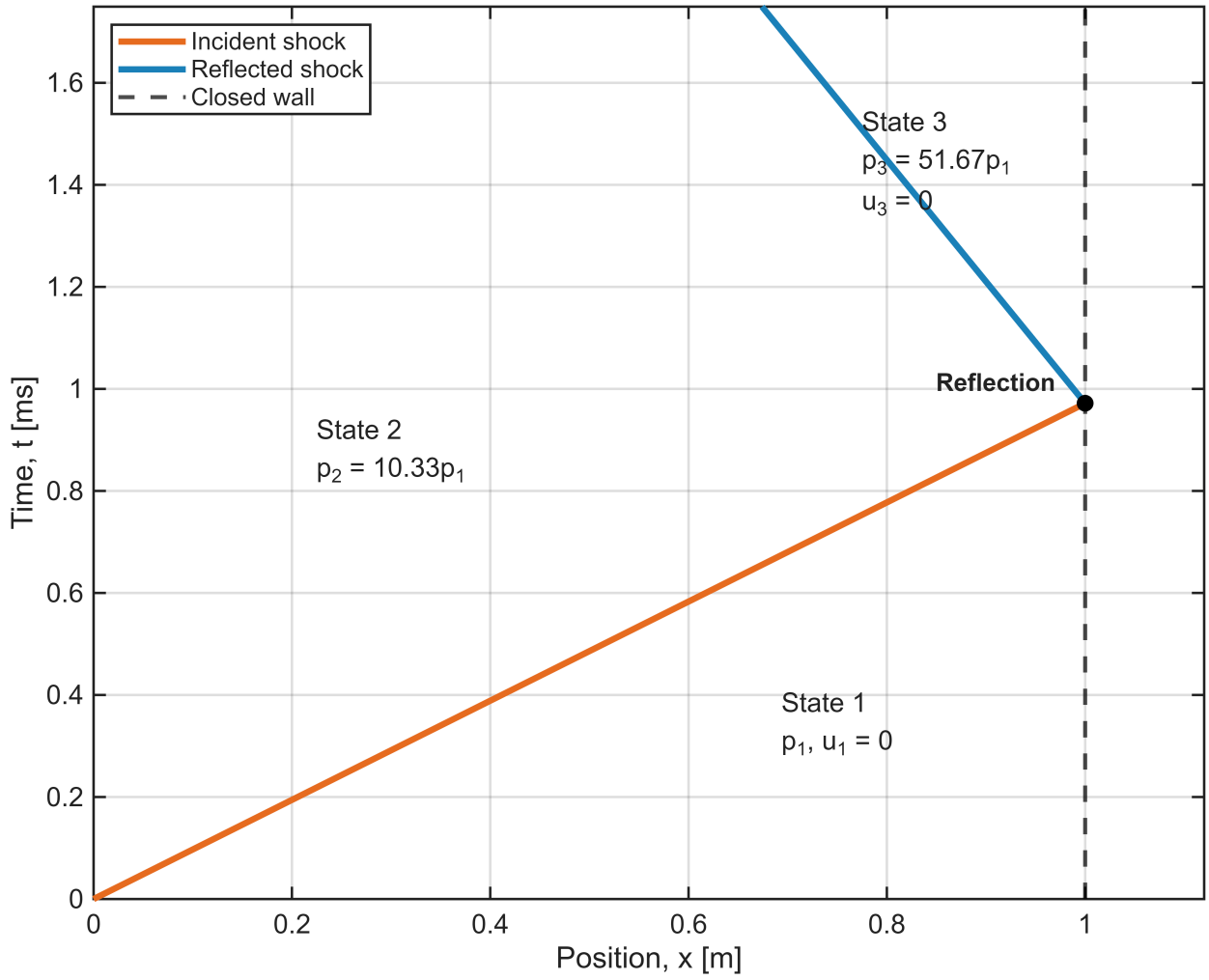
title('Mach 3 Shock Reflection at a Closed Wall', ...
    'FontSize',15,'FontWeight','bold');

legend([h1 h2 h3], ...
    {'Incident shock','Reflected shock','Closed wall'}, ...
```

```
    'Location','northwest');

xlim([0 1.12*L]);
ylim([0 1e3*tEnd]);

grid on;
set(gca,'FontSize',12,'LineWidth',1.1, ...
    'Color','w','GridAlpha',0.15);
```




```
%% 7. Print results
fprintf('\n--- Shock Reflection Results ---\n');
```

```
--- Shock Reflection Results ---
```

```
fprintf('Incident shock speed: %.2f m/s\n',Vs);
```

```
Incident shock speed: 1029.00 m/s
```

```
fprintf('Reflected shock speed: %.2f m/s\n',Vr);
```

```
Reflected shock speed: -419.22 m/s
```

```
fprintf('Reflection time:       %.4f ms\n',1e3*tR);
```

```
Reflection time:       0.9718 ms
```

```
fprintf('Incident pressure:     %.4f p1\n',p2_p1);
```

```
Incident pressure:     10.3333 p1
```

```
fprintf('Wall pressure:         %.4f p1\n',p3_p1);
```

```
Wall pressure:         51.6667 p1
```

```
%% 8. Optional PDF export
% exportgraphics(gcf,'Mach3_Shock_Reflection.pdf', ...
%     'ContentType','vector');
```

Problem 2. Acoustic Wave Propagation

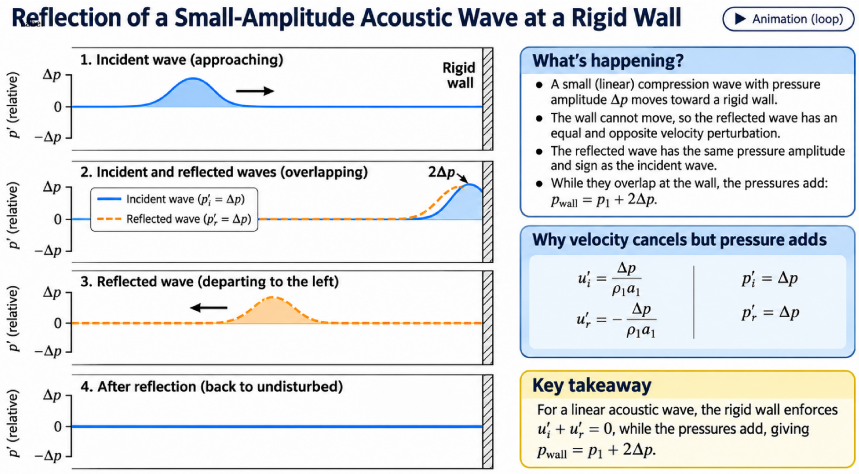
Problem Statement

Suppose that the initial wave in Problem (1) is instead acoustic, of small pressure amplitude . What is the pressure on the end wall after the wave arrives? Explain.

Analysis

1. Understand the physical situation

```
%imshow('Acoustic Wave Reflection at a Rigid Wall.png')
I = imread('Acoustic Wave Reflection at a Rigid Wall.png');       % original image
position = [50 30];                 % [x y]
txt = 'Label';
RGB = insertText(I, position, txt, FontSize=18, BoxOpacity=0);  % returns truecolor
figure; imshow(RGB, 'InitialMagnification', 'fit');
```



→ A small-amplitude acoustic compression wave travels to the right through initially stationary gas.

Define:

- p_1: Initial, undisturbed gas pressure.
- p': Small pressure increase associated with the incident acoustic wave.
- a_1: Speed of sound in the undisturbed gas.
- u': Small fluid velocity induced by the wave.

Assume:

$$\frac{p'}{p_1} << 1$$

(Condensation $s << 1$, allows the use of linear acoustic theory; i.e., the pressure of the undisturbed gas is much greater than the pressure increase associated with the acoustic wave.)

2. Apply the boundary condition at the wall.

$$u_{wall} = 0$$

The incident acoustic wave induces a small positive fluid velocity u_i', directed to the right.

The reflected wave must produce an equal and opposite velocity disturbance:

$$u_r' = -u_i'$$

3. Calculate the pressure on the end of the wall after the wave arrives.

Use the linear acoustic pressure-velocity relationship.

For a small-amplitude plane acoustic wave traveling to the right,

$$p_i' = \rho_1 a_1 u_i'$$

For an acoustic wave traveling to the left, the relationship becomes

$$p_r' = -\rho_1 a_1 u_r'$$

Substituting the velocity boundary condition gives:

$$p_r' = \rho_1 a_1 u_i'$$

- The pressure disturbance caused by the reflected acoustic wave matches that of the incident wave.

The total pressure on the wall is the undisturbed pressure plus the two disturbances:

$$p_{\text{wall}} = p_1 + p_i' + p_r'$$

Since

$$p_r' = p_i' = p'$$

we obtain

$$p_{\text{wall}} = p_1 + 2p'$$

For a small pressure increase associated with the incident acoustic wave, $p' = \Delta p$, therefore:

$$p_{\text{wall}} = p_1 + 2\Delta p$$

Here, Δp is the incident acoustic pressure increase above the initial pressure p_1.

Physical interpretation:

When a small-amplitude acoustic compression wave reaches a rigid wall, the zero-velocity boundary condition requires an equal-amplitude reflected wave traveling in the opposite direction. Although their induced particle velocities cancel, their pressure disturbances reinforce each other. Therefore, the pressure increase at the wall is twice the incident pressure increase.

x-t diagram:

```
%% Acoustic Wave Reflection at a Rigid Wall
% Compressible Fluid Mechanics - Problem 2
% Clean x-t diagram with separate wave colors and annotations

clear; clc; close all;

%% 1. Parameters
```

```
a1 = 343;                    % Speed of sound [m/s]
L  = 1.0;                    % Tube length [m]

tR   = L/a1;                 % Reflection time [s]
tEnd = 2*tR;                 % Full reflected trajectory

% Colors
blue   = [0.10 0.40 0.75]; % Incident wave
orange = [0.90 0.40 0.10]; % Reflected wave
gray   = [0.45 0.45 0.45];

%% 2. Wave trajectories
tIncident  = linspace(0,tR,300);
xIncident  = a1*tIncident;

tReflected = linspace(tR,tEnd,300);
xReflected = L-a1*(tReflected-tR);

% Convert time to milliseconds
tI_ms = 1e3*tIncident;
tR_ms = 1e3*tReflected;
tWall_ms = 1e3*tR;

%% 3. Create figure
fig = figure('Color','w', ...
    'Position',[150 100 1000 750]);

ax = axes(fig);
hold(ax,'on');
box(ax,'on');

% Incident wave
h1 = plot(xIncident,tI_ms, ...
    'Color',blue,'LineWidth',3);

% Reflected wave
h2 = plot(xReflected,tR_ms, ...
    'Color',orange,'LineWidth',3);

% Rigid wall
h3 = plot([L L],[0 1e3*tEnd], ...
    '--','Color',gray,'LineWidth',2);

% Reflection point
plot(L,tWall_ms,'ko', ...
    'MarkerFaceColor','k','MarkerSize',7, ...
    'HandleVisibility','off');

%% 4. Region labels
% Place region labels away from both wave trajectories
```

```
text(0.75*L,0.27*tWall_ms, ...
    {'Undisturbed','gas'}, ...
    'HorizontalAlignment','center', ...
    'FontSize',13,'Color',[0.20 0.20 0.20]);

text(0.43*L,1.00*tWall_ms, ...
    {'Region behind','incident wave'}, ...
    'HorizontalAlignment','center', ...
    'FontSize',13,'Color',[0.20 0.20 0.20]);

text(0.75*L,1.65*tWall_ms, ...
    {'Region behind','reflected wave'}, ...
    'HorizontalAlignment','center', ...
    'FontSize',13,'Color',[0.20 0.20 0.20]);

%% 5. Wave call-outs with leader arrows
% The arrows are in data coordinates using quiver().
% Their labels are horizontal and separated from region text.

% Incident wave call-out
text(0.08*L,0.53*tWall_ms, ...
    'Incident acoustic wave', ...
    'FontSize',12,'FontWeight','bold', ...
    'Color',blue);

% Arrow points to incident wave at x = 0.48 L
quiver(0.30*L,0.50*tWall_ms, ...
       0.18*L,-0.02*tWall_ms,0, ...
       'Color',blue,'LineWidth',1.4, ...
       'MaxHeadSize',0.8, ...
       'HandleVisibility','off');

% Reflected wave call-out
text(0.07*L,1.42*tWall_ms, ...
    'Reflected acoustic wave', ...
    'FontSize',12,'FontWeight','bold', ...
    'Color',orange);

% Arrow points to reflected wave at x = 0.48 L
quiver(0.32*L,1.45*tWall_ms, ...
       0.16*L,0.07*tWall_ms,0, ...
       'Color',orange,'LineWidth',1.4, ...
       'MaxHeadSize',0.8, ...
       'HandleVisibility','off');

%% 6. Wall annotation
text(L+0.04*L,1.70*tWall_ms, ...
    {'Rigid wall','u = 0'}, ...
    'FontSize',12, ...
```

```
    'FontWeight','bold', ...
    'Color',gray, ...
    'HorizontalAlignment','left');

text(L+0.04*L,1.04*tWall_ms, ...
    'Reflection', ...
    'FontSize',12, ...
    'FontWeight','bold', ...
    'HorizontalAlignment','left');

%% 7. Axes formatting
xlabel('Position, x [m]','FontSize',14);
ylabel('Time, t [ms]','FontSize',14);

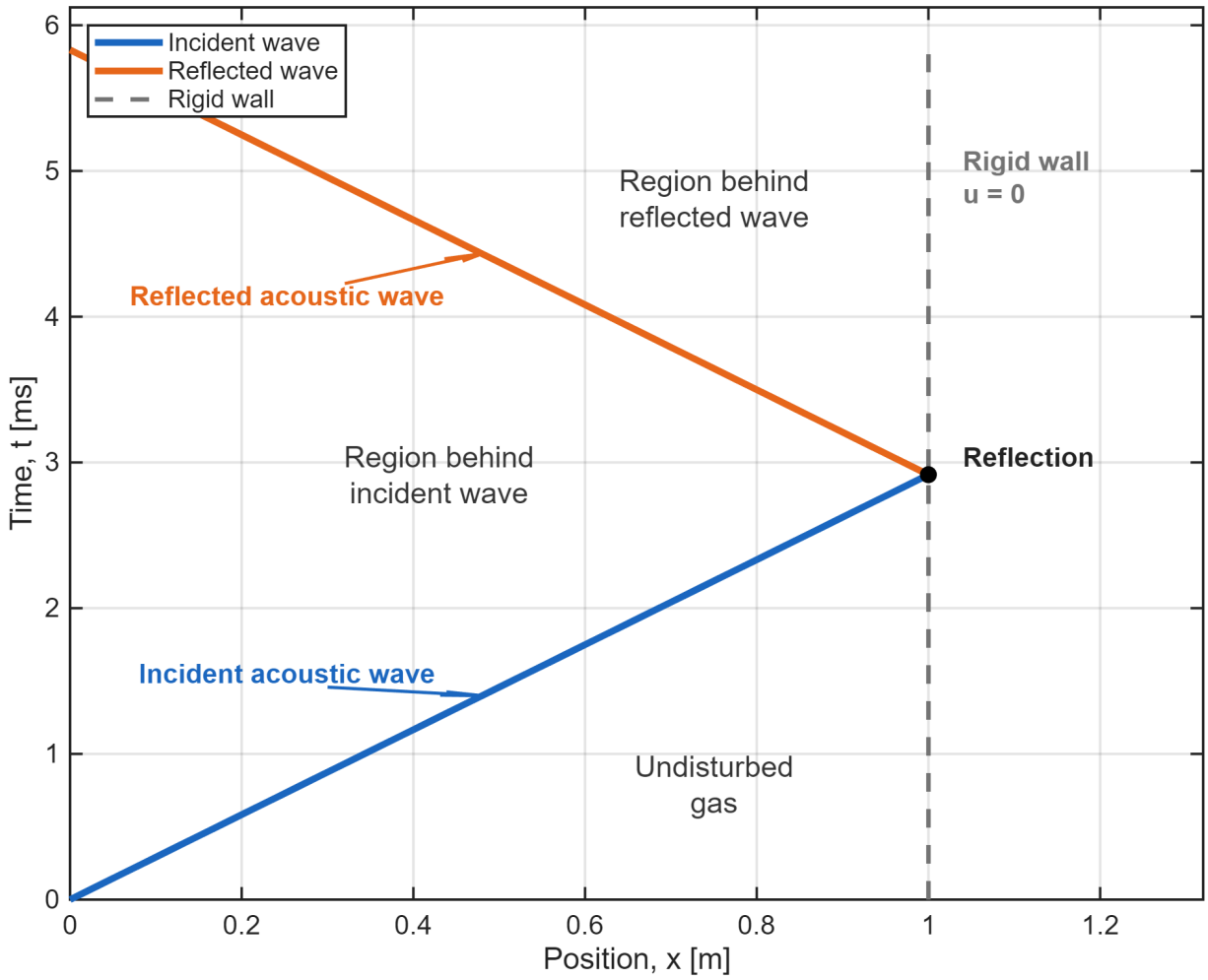
title('Reflection of a Small-Amplitude Acoustic Wave', ...
    'FontSize',16,'FontWeight','bold');

xlim([0 1.32*L]);
ylim([0 2.10*tWall_ms]);

legend([h1 h2 h3], ...
    {'Incident wave','Reflected wave','Rigid wall'}, ...
    'Location','northwest', ...
    'FontSize',11);

set(gca, ...
    'FontSize',12, ...
    'LineWidth',1.1, ...
    'Color','w', ...
    'GridAlpha',0.12);

grid on;
```




```
%% 8. Print results
fprintf('\n--- Acoustic Wave Reflection ---\n');
```

```
--- Acoustic Wave Reflection ---
```

```
fprintf('Sound speed:              %.2f m/s\n',a1);
```

```
Sound speed:              343.00 m/s
```

```
fprintf('Reflection time:          %.4f ms\n',tWall_ms);
```

```
Reflection time:          2.9155 ms
```

```
fprintf('Return time to x = 0:     %.4f ms\n',1e3*tEnd);
```

```
Return time to x = 0:     5.8309 ms
```

```
fprintf('Wall pressure: p_wall = p1 + 2*Delta_p\n');
```

```
Wall pressure: p_wall = p1 + 2*Delta_p
```

```
%% 9. Optional PDF export
% exportgraphics(fig, ...
%     'Acoustic_Wave_Reflection_xt.pdf', ...
%     'ContentType','vector');
```

Unlike our Mach 3 shock problem, the small-amplitude acoustic waves propagate at approximately the same sound speed a_1 in opposite directions.

Problem 3. Pressure Fields and Fluid Acceleration

Problem Statement

Based off of the movie "Pressure Fields and Fluid Acceleration" [3], write one paragraph aimed at a "bright high school kid" that explains one aspect of physics illustrated in the movie.

Statement

Imagine water flowing toward the nose of an object (a submarine, or a ship!): most of the flow turns and flows around the sides, but right at the nose, the water comes to a stop. As it slows down, its pressure rises. **Stagnation pressure** is the pressure a moving fluid would reach if it were brought to rest (to a complete stop!) without losses from friction or other irreversible effects.

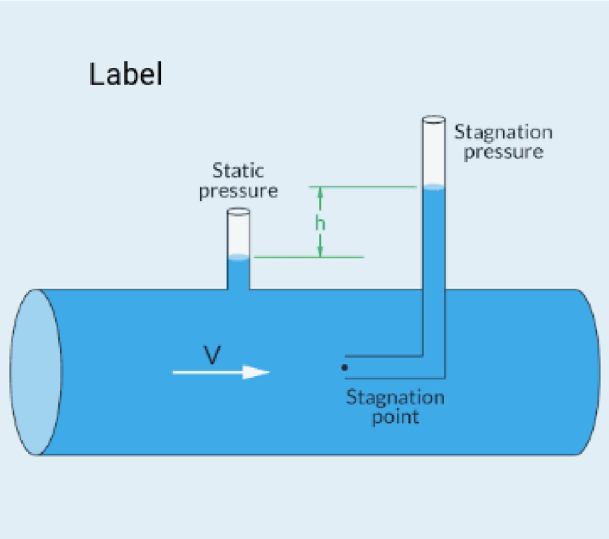
In the movie's nearly frictionless water experiments, it equals the fluid's **static** (or "ordinary") **pressure** plus its **dynamic pressure**, $(\frac{1}{2}\rho u^2)$, which represents its kinetic energy per unit volume.

$$p + \frac{1}{2}\rho u^2 = p_{stagnation}$$

As water speeds up through a contraction, its static pressure falls, but its stagnation pressure stays approximately constant. A Pitot tube measures stagnation pressure by bringing the fluid to rest at its forward-facing opening; comparing that reading with the static pressure lets us calculate the flow speed.

```
%imshow("pitot_tube.png");

I = imread('pitot_tube.png');       % original image
position = [50 30];                 % [x y]
txt = 'Label';
RGB = insertText(I, position, txt, FontSize=18, BoxOpacity=0);  % returns truecolor
figure; imshow(RGB, 'InitialMagnification', 'fit');              % new figure, no
hold
```



References

[1] Liepmann, H. W., & Roshko, A. (2001). *Elements of Gasdynamics*. Dover Publications. (Original work published 1957)

[2] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.

[3] YouTube '6. Pressure Fields and Fluid Acceleration'.